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CTF101

File Formats :-

File signatures(Magic bytes)

Bytes within a file to determine the type of the file.

The tool used to know the file signatures is xxd.

Meta Data :-

Meta Data :- Basically Data about Data.

For Ex :- Meta Data about an Image can include GPS Location, Date when it is captured, etc.

The tool used to know the meta data is exiftool.

TimeStamps :-

Indicate the time of certain events(MAC)

M(Modified), A(Accessed), C(Created)

There are patterns in timestamps :-

Pattern: Run steghide Tool on File

If the steghide tool was used to hide information in a file,

M= Date Changed (MFT) = INDX Entry Date Modified = INDX Entry Date Changed > A, C



Pattern: File Renamed

If a file was renamed,

Date Changed (MFT) $\neq$ INDX Entry Date Changed and Date Changed (MFT) > M, A, C



Pattern: File Copied Within Drive

If a file was copied within a drive,

A = C > Date Changed (MFT), Filename Date Changed (MFT)

Copy's Filename Date Modified (MFT) = Orig's Filename Date Modified (MFT)

Copy's Filename Date Changed (MFT) = Orig's Date Changed (MFT)



Pattern: File Copied from Different Drive

If a file was copied from a completely different drive,

Accessed = Created > Modified

This is because the file must be newly created & accessed when copying the data over, but the last time it was modified was from before this time.



Pattern: File Opened with Paint, No Save

If the steghide tool was used to hide information in a file,

Filename Date Changed (MFT) > M, A, C



MFT(Master File Table) :-

Think of the MFT as a hidden, behind-the-scenes ledger that Windows uses to keep track of every file. While regular users only see the MAC times (Modified, Accessed, Created), forensic investigators look at the secret MFT timestamps to see if a file's background records have been altered.

Steganography :-

To hide the data in plain sight.

Often embedded in images or audio.

How we Find something is hidden in a image or something :-

Using Timestamps if it is modified let's assume yesterday but created tomorrow, there is chances that something is hidden.

Core Technique(LSB)

Imagine you are buying a house for \$1,000,000. If the price changes to \$1,000,001, nobody cares. That single \$1 is the "Least Significant" part of the number.

How can someone hide something in an image using this technique?

In computer language (binary), the number 255 looks like **11111111** (pure white).

If you change the very last digit (the Least Significant Bit) to a zero, the code becomes **11111110** (the number 254).

Doesn't matter right?

Disk Imaging :-

Digital forensics works the exact same way. If you plug a suspect's USB drive into your computer and start clicking around, your computer will automatically create hidden files, update Accessed timestamps, and taint the evidence.

To solve this, investigators use Disk Imaging. A forensic image is a massive, exact, bit-by-bit digital clone of a hard drive or USB. Once the clone is created, the investigator locks the original drive in an evidence locker and only does their detective work on the copy.

1.Checksum

Programs like **md5sum** or **sha256sum** scan every single 1 and 0 on a hard drive and do a massive math equation to spit out a string of letters and numbers (e.g., **8d2645...**).

If even a *single letter* in a hidden text document is changed, the math equation breaks, and the resulting barcode changes completely

2. WriteBlocker

It's a tool.(One way)

It allows your computer to *read* the data on the suspect's drive to copy it, but it physically blocks your computer from *writing* or saving any new data back onto it.

Kali Linux has mode 'Forensics' that features WriteBlocker.

Memory Forensics :-

When a computer is turned on and actively running programs, it stores its temporary, real-time data in RAM.

The problem with RAM is that it is volatile—the second the computer is turned off or restarted, *everything* in RAM is wiped clean forever. Memory Forensics is the process of taking a snapshot of this active RAM (called a "memory dump") before the computer turns off, so investigators can see exactly what the computer was doing at that exact second.

These dumps of data are often very large, but can be analyzed using a tool called Volatility.

There should be a good Work Flow when using Volatility.

Hex Editors:

Use Hexcurse(Tool) to correct the File Signatures or Magic Bytes when File signatures are wrong or someone knowingly changed it.